

Aim : To apply ML algorithm for predictive maintenance

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```
In [68]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.tree import plot_tree
from sklearn.metrics import accuracy_score
```

```
In [69]: ap=pd.read_csv(r"C:\Users\Admin\Downloads\predictive_maintenance.csv")
```

```
In [70]: ap.head()
```

```
Out[70]:
```

	Temperature	Vibration	Pressure	Maintenance_Required
0	88	24	144	1
1	78	41	129	1
2	64	41	141	1
3	92	33	105	1
4	57	21	102	0

```
In [71]: ap.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Temperature           200 non-null   int64
1   Vibration              200 non-null   int64
2   Pressure               200 non-null   int64
3   Maintenance_Required  200 non-null   int64
dtypes: int64(4)
memory usage: 6.4 KB
```

```
In [72]: ap.describe()
```

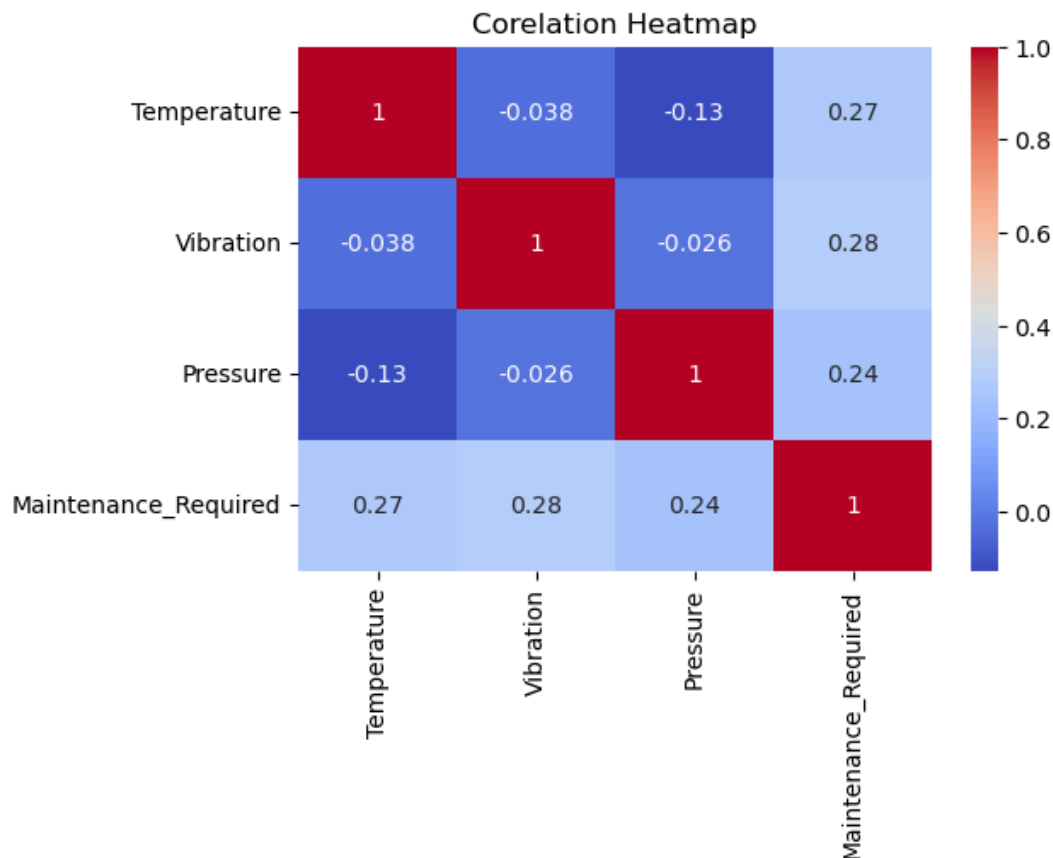
```
Out[72]:
```

	Temperature	Vibration	Pressure	Maintenance_Required
count	200.000000	200.000000	200.000000	200.000000
mean	74.260000	30.690000	119.935000	0.885000
std	14.229716	11.797875	16.732323	0.319823
min	50.000000	10.000000	90.000000	0.000000
25%	62.000000	20.750000	106.000000	1.000000
50%	75.000000	32.000000	119.500000	1.000000
75%	87.250000	41.000000	135.250000	1.000000
max	99.000000	49.000000	149.000000	1.000000

```
In [73]: ap.shape
```

```
Out[73]: (200, 4)
```

```
In [74]: plt.figure(figsize=(6,4))
sns.heatmap(ap.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



```
In [78]: # Feature Selection
# Input & Output

x=ap[['Temperature', 'Vibration', 'Pressure']]
y=ap['Maintenance_Required']
```

```
In [80]: # Split Data
x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.2, random_state=42)
```

```
In [81]: # Create Decision Tree model
model= DecisionTreeClassifier()
```

```
In [82]: # Train Model
model.fit(x_train, y_train)
```

```
Out[82]: ▾ DecisionTreeClassifier ⓘ ?
          ► Parameters
```

```
In [83]: # Prediction
y_pred=model.predict(x_test)
```

```
In [85]: # Accuracy
accuracy= accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy)
```

Model Accuracy: 1.0

```
In [89]: # New data
new_data = [[120, 85,15]]
prediction=model.predict(new_data)
```

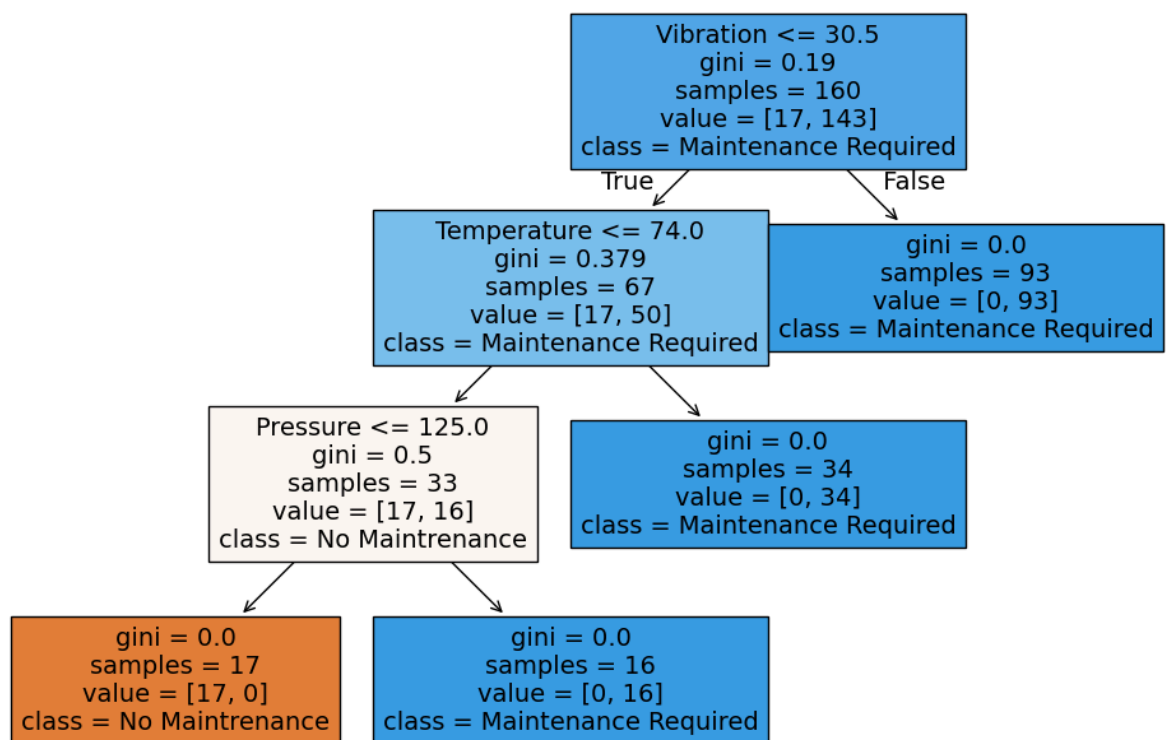
```
# output
if prediction[0] ==1:
    print("Maintenance Requiried")
else:
    print("NO Maintenance Requires")
```

Maintenance Requiried

C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
warnings.warn(

```
In [90]: # Display Decision Tree
plt.figure(figsize=(12,8))
plot_tree(
    model,
    feature_names=['Temperature', 'Vibration', 'Pressure'],
    class_names=['No Maintenance', 'Maintenance Required'],
    filled=True
)
plt.title(" Decision Tree For Predictive Maintenance")
plt.show()
```

Decision Tree For Predictive Maintenance



In []: